MISSING DATA MECHANISMS: AN INTERACTIVE MONTE CARLO DASHBOARD FOR LONGITUDINAL CLINICAL TRIALS

By

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Abstract

Lead exposure in children presents a serious health concern, primarily resulting from the inhalation or ingestion of lead dust from decaying paint in homes built before 1978. Exposure causes permanent damage to the brain and nervous system, impacting development and leading to learning and behavioral problems, fatigue, and poor appetite. The most effective method to detect exposure is a blood test, as symptoms can range from subtle indicators, such as irritability and lack of focus, to severe reactions like vomiting and seizures (Centers for Disease Control and Prevention, 2022).

The goal of this project is to build an interactive dashboard that illustrates missing data mechanisms using clinical trial data as a framework. By simulating differential attrition, this tool evaluates the effect missing data has on longitudinal clinical trials, quantifying how it impacts statistical significance and the validity of conclusions drawn from studies with incomplete data.

The interactive Shiny dashboard accompanying this thesis is hosted online and can be accessed via two platforms. For internal review and university faculty, the tool is deployed on the institutional server at: https://viz.datascience.arizona.edu/tlc_shiny_dashboard/. For external access and public viewing, a mirrored deployment is available at: https://jennyygeng.shinyapps.io/TLC_Shiny_Dashboard/.

Motivating Application: The TLC Trial

The Treatment of Lead-Exposed Children (TLC) trial serves as the motivating case study for this simulation framework. The TLC trial was a randomized, placebo-controlled study of succimer in children with moderate blood lead levels (20-44 μg/dL). Succimer (DMSA) is an oral chelating agent—a medication that binds to metal ions in the body to prevent harm and facilitate excretion—used to treat heavy metal poisoning, specifically lead poisoning in children. The study investigated succimer's effects on 780 toddlers (aged 12-33 months) from high-risk housing, who were predominantly African-American. The trial found that while succimer effectively lowered blood lead levels temporarily, the effects were short-lived, with lead levels beginning to rebound by the end of the 6-week follow-up period (TLC Trial Group, 2000). Furthermore, long-term follow-up studies demonstrated no cognitive or behavioral benefits and were even associated with slightly worsened behavior. These findings led to the conclusion that chelation with succimer is not beneficial for cognitive or behavioral issues at levels below 45 μg/dL, shifting the focus from treating low-level exposure to preventing lead exposure in the first place (Rogan et al., 2001).

The data utilized for this project consists of four repeated measurements of blood lead levels obtained at baseline (Week 0), Week 1, Week 4, and Week 6. These measurements were taken from a subsample of 100 children who were randomly assigned to either the chelation treatment with succimer (Active) or a placebo. Additionally, simulated values for age and sex have been appended to the original dataset. It is important to note that while the toddlers in the original trial were between 12 and 33 months old, the simulated age values extend beyond these limits.

A comprehensive baseline summary of the simulated TLC study cohort is presented in Table 1.

Table 1: **Blood Lead Level Summary by Treatment Group**

Variable	**Overall**(N=100)[1]	**Active**(N=50)[1]	**Placebo**(N=50)[1]	**p-value**[2]
Sex				0.008
female	41 (41%)	27 (54%)	14 (28%)	
male	59 (59%)	23 (46%)	36 (72%)	
Age (months)	24.9 (6.5 - 38.6)	25.3 (11.2 - 38.6)	24.4 (6.5 - 34.9)	0.3
Week 0 (Baseline)	25.6 (19.7 - 41.1)	25.3 (19.7 - 38.1)	26.2 (19.7 - 41.1)	0.8
Week 1	21 (3 - 41)	24 (15 - 41)	12 (3 - 39)	<0.001
Week 4	20 (3 - 40)	22 (15 - 39)	15 (3 - 40)	<0.001
Week 6	21 (4 - 64)	22 (14 - 43)	19 (4 - 64)	0.006

[1] n (%); Median (Min - Max)
[2] Pearson's Chi-squared test; Wilcoxon rank sum test

The summary table classifies the data by treatment assignment, Active (Succimer) and Placebo, while also providing an overall column for the total sample. Continuous variables, such as age (measured in months) and blood lead levels, are summarized using the median and the range (minimum - maximum). This non-parametric summary was chosen over the mean and standard deviation to provide a robust measure of central tendency, as clinical data often exhibits skewed distributions or contains outliers that can distort the mean. The p-value column indicates the statistical significance of differences between the Active and Placebo groups at baseline. These values were calculated using Pearson's Chi-squared test for categorical variables (Sex) and the Wilcoxon rank-sum test for continuous variables.

To fully understand the specific naming conventions and data types used in the subsequent analyses, Table 2 provides a complete data dictionary for the variables summarized above.

Table 2: **Data Dictionary for the Simulated TLC Dataset**

Variable	Type	Description
ID	Integer	A unique identifier for each subject.
Group	Character	The treatment group: 'A' (Active) or 'P' (Placebo).
Week0	Numeric	Blood lead level at baseline (Week 0).
Week1	Numeric	Blood lead level after one week.
Week4	Numeric	Blood lead level after four weeks.
Week6	Numeric	Blood lead level after six weeks.
sex	Character	The biological sex of the subject.
age	Numeric	The age of the subject in months.

To understand the dynamics of lead levels throughout the trial, we did exploratory visualizations on the individual trajectories, group distributions at each time point, and the overall change from baseline to Week 6.

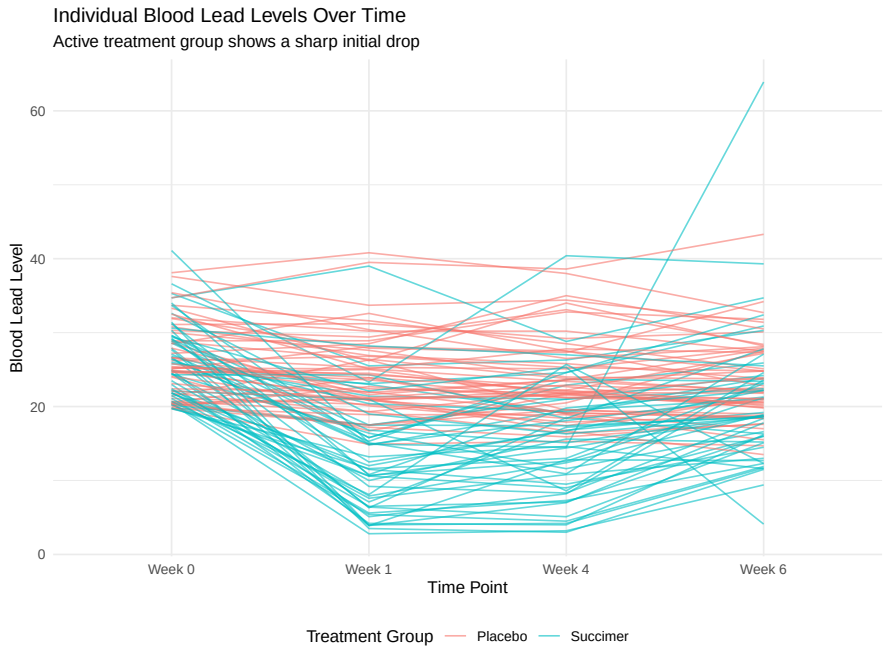

Figure 1. Individual Blood Lead Levels Over Time. A longitudinal "spaghetti plot" visualizing the individual response profiles of all subjects across the 6-week trial.

As seen in Figure 1, a distinct separation between the cohorts can be seen immediately. Subjects in the Active group (red) exhibit a sharp, dramatic decline in blood lead levels within the first week of treatment. Following this initial drop, many subjects in the Active group show a slight "rebound" or stabilization pattern in Weeks 4 and 6, rather than a continued sharp decline. In contrast, subjects in the Placebo group (green) show trajectories that are generally flat or decline very gradually, lacking the immediate treatment response seen in the Active group. One notable outlier in the Active group exhibits a sharp increase in lead levels by Week 6, diverging from the general group trend.

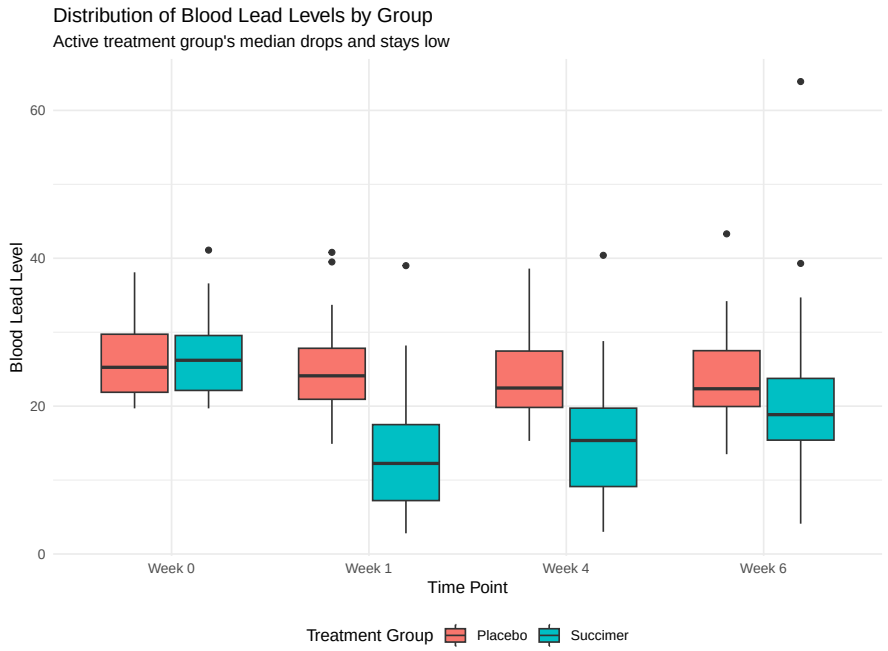

Figure 2. Distribution of Blood Lead Levels by Group. Boxplots displaying the median, interquartile ranges, and outliers for both treatment groups at each specific time point.

The boxplots in Figure 2 confirm the trends observed in the individual trajectories. At Week 0 (Baseline), the median blood lead levels and interquartile ranges (IQR) for both groups are highly comparable, indicating successful randomization. By Week 1, the median for the Active group drops significantly to approximately 12-15 mcg/dL, while the Placebo group remains near baseline levels (~25 mcg/dL). During Weeks 4 and 6, the Active group maintains lead levels lower than the Placebo group, but the median gradually rises after Week 1. This suggests that while the treatment is effective, blood lead levels may begin to creep back up after the initial chelation. The Placebo group shows a very slow, gradual decline but remains consistently higher than the Active group, with several outliers visible above the upper whiskers in both groups during the later weeks.

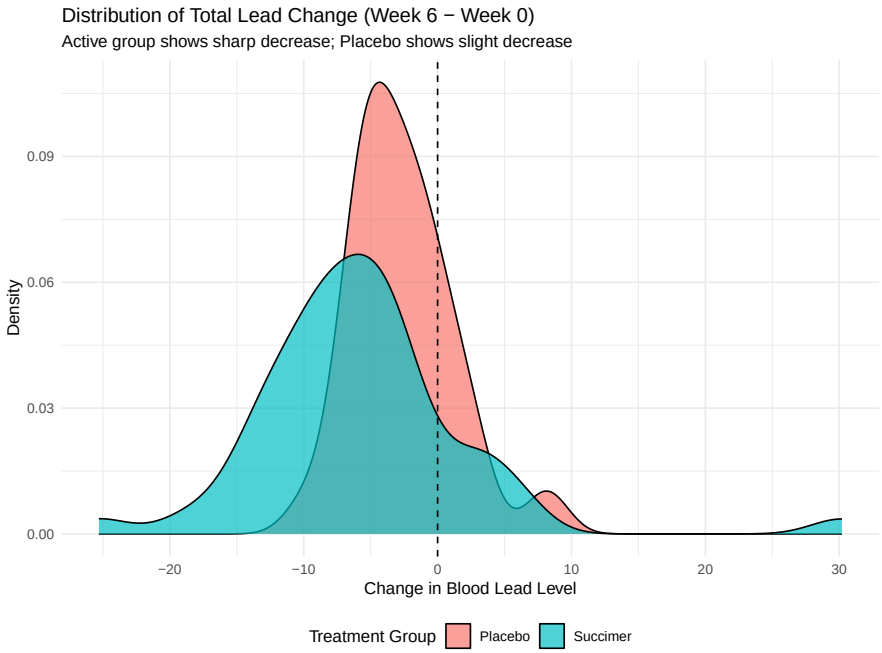

Figure 3. Distribution of Total Lead Change (Week 6 - Week 0). Density plots illustrating the overall change in blood lead levels from baseline to the final follow-up visit.

Examining the overall change distributions (Figure 3), the Active group's distribution is roughly bell-shaped but shifted heavily to the left, centered around a reduction of approximately -10 to -12 mcg/dL. This indicates that the vast majority of treated children experienced a clinically significant decrease in lead levels. Conversely, the Placebo group's distribution is narrower and centered just slightly below zero. While some placebo subjects experienced minor natural decreases, likely due to regression to the mean or environmental changes, the overall magnitude of change is negligible compared to the Active group.

Dashboard Design and Features

To bridge the gap between abstract statistical theory and practical application, an interactive Shiny dashboard was engineered alongside this report using the `flexdashboard` framework. The primary objective of this tool is to provide immediate, visual, and statistical feedback on how model assumptions interact with missing data mechanisms.

The user interface was designed to grant users dynamic control over the "laws of physics" governing the simulated clinical trials. The core interactive components include:

- **Mechanism Toggle:** Allows the user to dictate the attrition logic, switching between Missing Completely at Random (MCAR) and covariate-dependent Missing at Random (MAR).
- **Missingness Proportion:** A slider controlling the total percentage of data lost during follow-up visits (ranging from 10% to 40%), enabling sensitivity analyses of trial degradation.

- **Number of Trials:** A control to specify the volume of parallel Monte Carlo simulations executed per run, ranging from 100 to 1,000 trials.
- **Covariate Correction Toggle:** A binary control that dictates whether the underlying statistical model ignores or accounts for baseline covariates (Age).

The dashboard's output relies heavily on reactive rendering. Upon execution, the tool generates a Monte Carlo distribution histogram mapping the simulated treatment effects against the true parameter. It dynamically calculates and displays the bias, empirical standard deviation, true estimate, and the mean of the simulated estimates. Additionally, the interface incorporates the explicit mathematical model being evaluated, accompanied by conditional text blocks that explain *why* the bias increased or decreased based on the user's specific parameter combinations.

Statistical Framework

To analyze the longitudinal trajectories of blood lead levels and evaluate the impact of missing data, we employed a unified framework centered around Linear Mixed-Effects Models (LME) and Monte Carlo simulations.

The Mixed-Effects Model The LME approach is superior to repeated measures ANOVA for this dataset as it explicitly models the correlation between repeated measurements within the same subject, handles unbalanced data seamlessly, and allows for the adjustment of baseline covariates. The full, adjusted model specifies blood lead level (Y_{ij}) for subject i at time j as:

$$\begin{aligned} Y_{ij} = \beta_0 &+ \beta_1 \text{Trt}_i + \beta_2 \text{Week1}_{ij} + \beta_3 \text{Week4}_{ij} + \beta_4 \text{Week6}_{ij} \\ &+ \beta_5(\text{Trt}_i \times \text{Week1}_{ij}) + \beta_6(\text{Trt}_i \times \text{Week4}_{ij}) + \beta_7(\text{Trt}_i \times \text{Week6}_{ij}) \\ &+ \beta_8 \text{Age}_i^c + \beta_9 \text{Sex}_i + b_{0i} + \epsilon_{ij}, \end{aligned}$$

where:

- Y_{ij}: Blood lead level for subject i at time j
- Trt_i: Indicator for Treatment group (0 = Placebo, 1 = Succimer)
- Week_{ij}: Indicator variables for time points (Reference = Week 0)
- Age_i^c: Centered age covariate for subject i ($\text{Age}_i - 22$)
- b_{0i}: Random intercept for subject i, assumed $b_{0i} \sim N(0, \sigma_b^2)$
- ϵ_{ij}: Residual error, assumed $\epsilon_i \sim N(0, \mathbf{R}_i)$.

Longitudinal clinical data frequently violates standard assumptions of independence and homoscedasticity. To address this, we modeled the residual variance as distinct for each time point ($\text{Var}(\epsilon_{ij}) = \sigma_j^2$) using the `varIdent` variance function, and modeled the correlation between repeated measures using a general symmetric correlation structure (`corSymm`). Model diagnostics confirmed the validity of these assumptions, with standardized residual and Q-Q plots showing no distinct patterns or curvature (see Appendix B).

The Data-Generating Mechanism (DGM) To systematically quantify the impact of missing data on this model's ability to detect treatment effects, we performed a Monte Carlo simulation. Rather than resampling a single static dataset, we utilized a complete DGM to build a "synthetic universe" of 100 brand-new subjects for every trial. Subjects were assigned ages ranging from 6.5 to 38.6 months to match the TLC dataset. The true mean blood lead levels were hard-coded to exactly match the observed marginal means from our Exploratory Data Analysis, establishing a "True" Treatment Effect at Week 6 of exactly $-2.8 \mu g/dL$.

The true data-generating model driving the simulated universe is formally defined as:

$$Y_{ij} = \mu_{\text{Group, Week}} - 0.5(\text{Age}_i^c) + b_{0i} + \epsilon_{ij}$$

where $\mu_{\text{Group, Week}}$ represents the hard-coded group means, Age_i^c is the subject's centered age, $b_{0i} \sim N(0, 4.42134^2)$ is the random intercept establishing within-subject correlation, and $\epsilon_{ij} \sim N(0, 2.460057^2)$ is the random measurement error.

Clinically, the random intercept standard deviation (4.42) represents how much children naturally differ from one another before the trial begins, while the residual standard deviation (2.46) represents random, day-to-day measurement error. Adding this variance ensures the simulated trajectories contain realistic clinical noise.

Crucially, the -0.5 coefficient establishes a strong, true age effect across all time points. While the empirical age coefficient in our provided dataset was highly weak (-0.051), we deliberately amplified it to -0.5 in our synthetic universe. This acts as a stress test for our mixed-effects models while reflecting the true clinical reality of environmental lead exposure: younger toddlers (12–24 months) are at the peak of crawling and hand-to-mouth behaviors, resulting in significantly higher rates of lead dust ingestion. To ensure the simulation remains computationally efficient, the models fitted during the simulation loops utilized a simplified Random Intercept model assuming homoscedasticity, which is mathematically sufficient to demonstrate the mechanisms of missing data bias.

Missing Data Mechanisms

Following data generation, missing data mechanisms were implemented with strict baseline protection. To maintain clinical validity, Week 0 data was protected from deletion, as baseline characteristics are recorded before treatment-related dropout can occur. Missing data was simulated exclusively during the follow-up visits (Weeks 1, 4, and 6) by deleting 20% of the total dataset using two distinct mechanisms:

Missing Completely at Random (MCAR) Under MCAR, every eligible post-baseline observation possesses an equal, uniform probability of being deleted. This serves as a control condition to demonstrate that random missingness reduces statistical power but does not inherently introduce systematic bias.

Missing at Random (MAR) Under MAR, the probability of missingness depends systematically on observed covariates. We simulated a realistic clinical attrition pattern where parents of younger children assigned to the Placebo group were highly likely to withdraw due to a perceived lack of improvement. The probability of missingness was weighted exponentially: $(40 - \text{Age})^3$ for the Placebo group, while the Succimer group retained a minimal baseline risk. This clinically mimics real-world dynamics: older placebo subjects naturally stabilize due to decreased hand-to-mouth behaviors, while treated subjects remain in the trial because the drug works. In contrast, the youngest placebo subjects exhibit stubbornly high lead levels, prompting early withdrawal to seek necessary medical treatment.

Missing Not at Random (MNAR) & Limitations It is important to note that this framework currently limits its scope to MCAR and MAR. MNAR occurs when missingness depends on the unobserved value itself (e.g., a child drops out *because* their lead level spiked dangerously high that specific week, before it could be measured). Accounting for MNAR requires highly complex joint-modeling techniques and remains a known limitation of the current simulation framework.

Demonstration: Simulation Results

Mixed-Effects Model applied to the TLC Trial When applying our adjusted mixed-effects framework to the complete motivating TLC dataset, we observe the following trajectories and statistical effects:

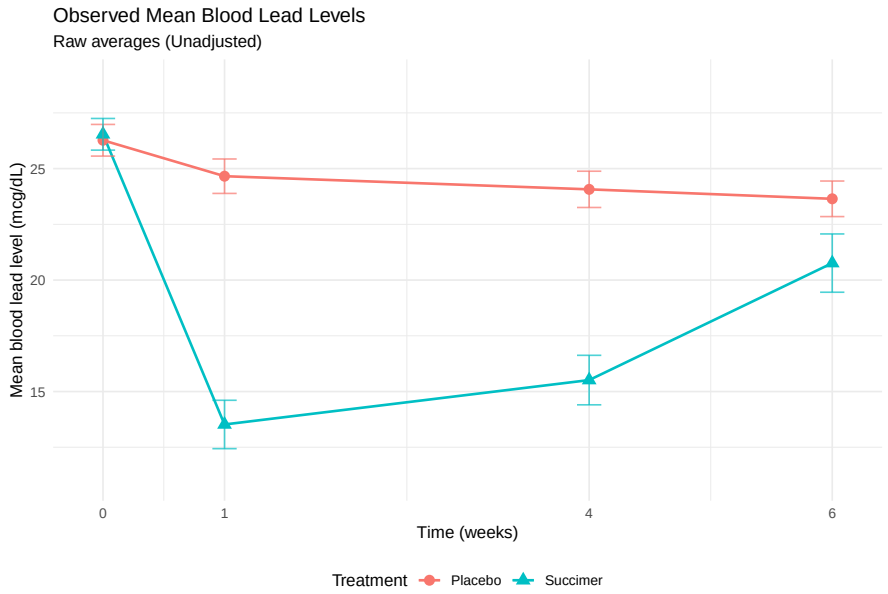

Figure 4. Observed Mean Blood Lead Levels. Raw empirical mean trajectories for the Succimer and Placebo groups. Error bars represent the standard error of the mean.

Figure 4 displays the raw empirical means of the trial data. Unadjusted for any demographic covariates, the Succimer group clearly exhibits a steep decline at Week 1, followed by a partial "rebound" pattern in subsequent weeks.

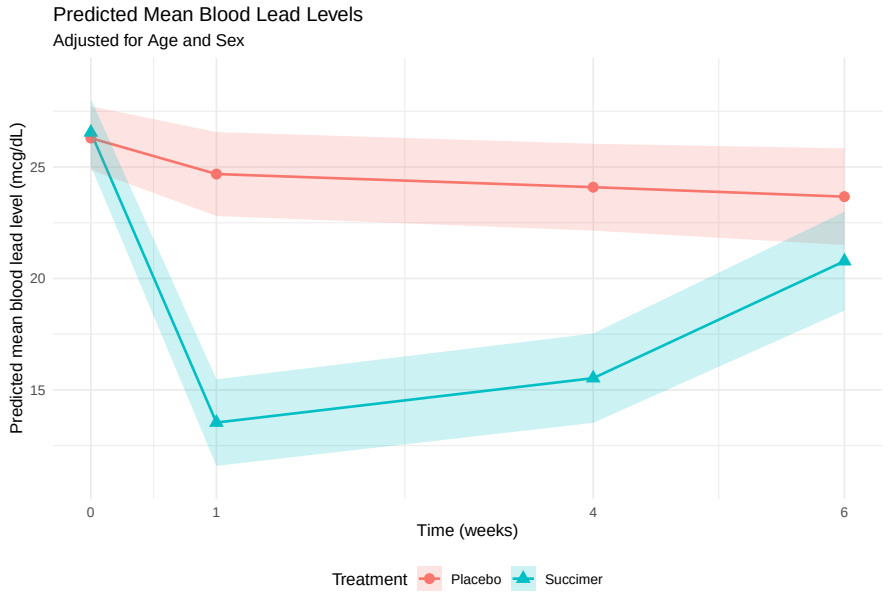

Figure 5. Predicted Mean Blood Lead Levels. Model-estimated marginal means adjusted for Age and Sex. Shaded regions represent 95% confidence intervals.

In contrast, Figure 5 shows the model-estimated marginal means. By explicitly including Age and Sex as covariates, the predicted model accounts for potential demographic imbalances, providing a more precise, mathematically isolated estimate of the true treatment effect.

Table 3: **Fixed Effects from the Adjusted Mixed-Effects Model**

	Value	Std.Error	DF	t-value	p-value
Intercept (Placebo, Wk 0)	27.673	2.191	294	12.628	0.000
Treatment: Succimer	0.256	1.060	96	0.241	0.810
Time: Week 1	-1.612	0.792	294	-2.036	0.043
Time: Week 4	-2.202	0.815	294	-2.702	0.007
Time: Week 6	-2.626	0.889	294	-2.955	0.003
Age (Months)	-0.051	0.083	96	-0.618	0.538
Sex: Male	-0.214	1.059	96	-0.202	0.840
Succimer x Week 1	-11.406	1.120	294	-10.184	0.000
Succimer x Week 4	-8.824	1.152	294	-7.657	0.000
Succimer x Week 6	-3.152	1.257	294	-2.508	0.013

The model confirms that both groups started at similar levels. There was no statistically significant difference in blood lead levels between the Placebo and Succimer groups at baseline (β_1, $p = 0.81$), confirming successful randomization. The interaction between Treatment and Week 1 was highly significant ($p < 0.001$), estimating that the Succimer group experienced a sharp divergence from the Placebo group immediately following the initiation of chelation. Significant interactions at Week 4 ($p < 0.001$) and Week 6 ($p = 0.013$) indicate that the treatment effect persisted throughout the study duration.

Table 4: **Post-Hoc Treatment Contrast at Week 6**

Contrast	Estimate	SE	Lower 95% CI	Upper 95% CI	p-value
Succimer vs. Placebo	-2.896	1.574	-6.02	0.228	0.069

Finally, by Week 6, the model estimates the Succimer group's mean lead level to be approximately 2.9 $\mu g/dL$ lower than the Placebo group (95% CI: -6.02 to 0.23). While the interaction remains significant in the main model ($p = 0.013$), the specific contrast at this final time point is marginally significant ($p = 0.069$). The confidence interval including 0 indicates that we cannot rule out the possibility of no difference at this specific time point. This reflects the "rebound" effect, where lead levels in the treated group began to rise after the initial drop, narrowing the gap between the two groups by the end of the study.

Monte Carlo Simulation Results Next, we evaluated the statistical models against the simulated missing data mechanisms. In the first stage, we evaluated the baseline MCAR scenario.

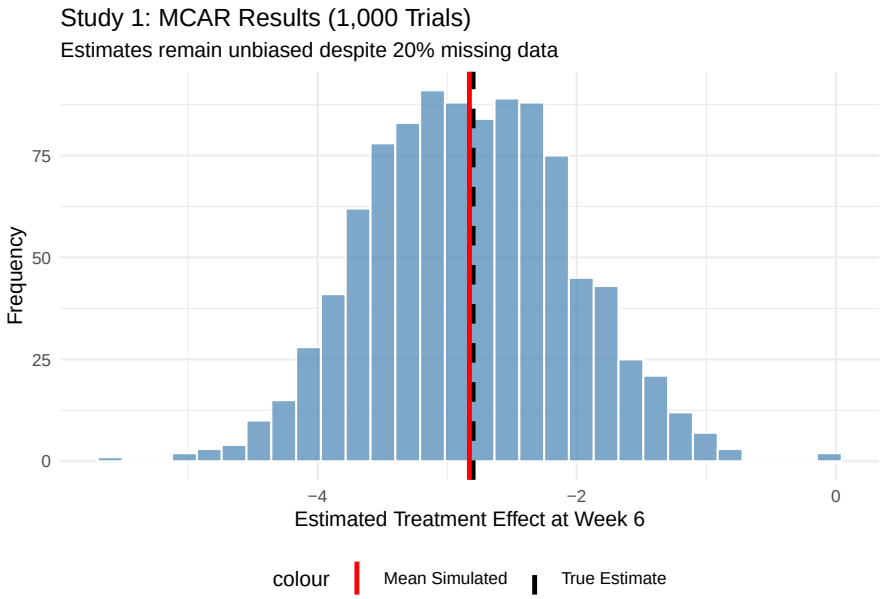

Figure 6. Simulation Results for MCAR Data. A histogram displaying the distribution of treatment effect estimates from 1,000 Monte Carlo simulation trials.

Table 5: **Simulation Statistics: MCAR**

Metric	Value
Mean Treatment Effect	-2.828
Bias	-0.028
Empirical Standard Deviation	0.775
CI Coverage (%)	95.400

As shown in Figure 6, the mean simulated estimate (solid red line) aligns almost perfectly with the True Estimate (black dashed line). The calculated bias is negligible (Bias = -0.028), confirming that when data is missing completely at random, the estimates remain mathematically unbiased.

In the second stage, we evaluated two mixed-effects models against the broken MAR dataset: an Unadjusted (Naive) model that ignores Age, and an Adjusted (Corrected) model that explicitly includes Centered Age as a covariate.

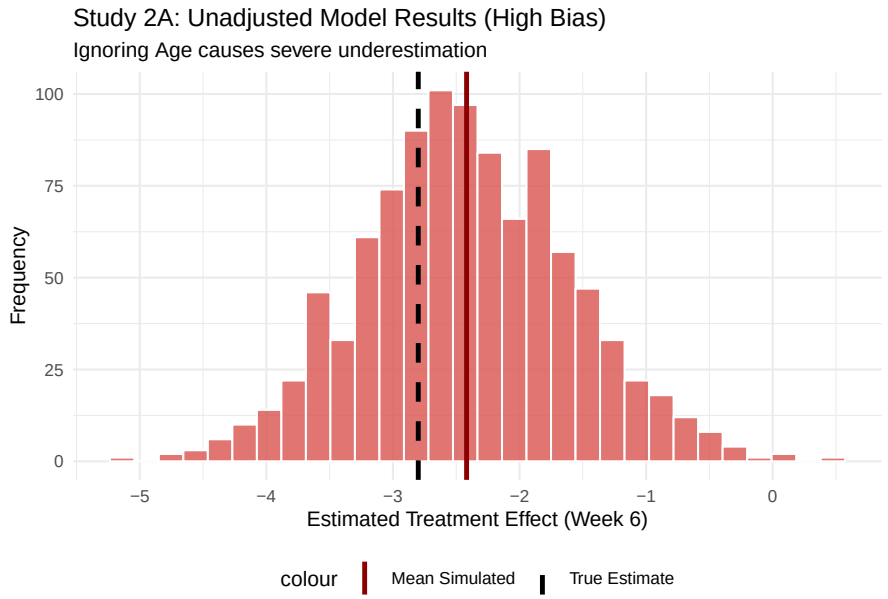

Figure 7. Unadjusted Model Results under Differential MAR. A histogram illustrating the biased distribution of treatment effect estimates from the naive statistical model.

Figure 7 demonstrates that the naive model (red) exhibits substantial bias because it compares the Succimer group to an artificially "healthier" subset of the Placebo group. By ignoring Age, the model fails to account for the systematic loss of younger, more severely affected children, causing it to severely underestimate the magnitude of the drug's true effectiveness (Bias = 0.382).

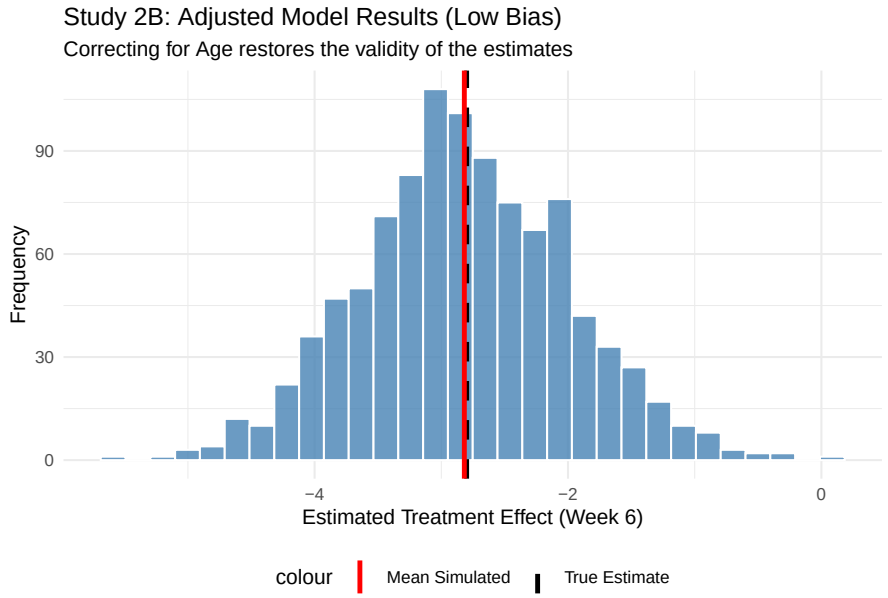

Figure 8. Adjusted Model Results under Differential MAR. A histogram illustrating the corrected distribution of treatment effect estimates from the age-adjusted model.

Table 6: **Simulation Statistics: MAR (Differential Dropout)**

Model	Mean Estimate	Bias	Empirical SD	Coverage (%)
Unadjusted (Naive)	-2.418	0.382	0.830	92.5
Adjusted (Corrected)	-2.818	-0.018	0.823	95.3

In contrast, the adjusted model (blue) in Figure 8 successfully demonstrates the mathematical power of covariate correction. Because the mixed-effects model utilizes Maximum Likelihood estimation and explicitly includes `age_c` as a covariate, it correctly identifies that the missing Placebo subjects were younger and therefore likely had higher lead levels. It mathematically recovers this missing information, completely neutralizing the bias and restoring the true estimate (Bias = -0.018).

Discussion and Limitations

This project successfully engineered a dynamic Monte Carlo simulation framework and interactive dashboard to quantify the impact of differential attrition in longitudinal clinical trials. By utilizing the TLC trial as a motivating case study, the tool demonstrated that while a linear mixed-effects model accurately captures treatment dynamics in complete datasets, its validity under missing data conditions is highly sensitive to model specification.

The simulation results conclusively show that MCAR missingness merely inflates standard errors, whereas MAR missingness—specifically, the systematic dropout of younger placebo subjects—introduces severe statistical bias. When the naive model failed to account for age, it artificially inflated the health of the Placebo group, drastically underestimating Succimer's efficacy. However, by explicitly incorporating Age as a covariate, the adjusted model leveraged Maximum Likelihood estimation to mathematically recover the missing information and neutralize the bias. These findings align strongly with established longitudinal methodologies (Fitzmaurice et al., 2011), confirming that researchers cannot rely on unadjusted analyses when attrition is linked to clinical or demographic factors.

A primary strength of this study is the rigor of the Data-Generating Mechanism, which spawns entirely new subjects for each trial rather than bootstrapping a static sample, effectively capturing true sampling variance. Additionally, the interactive nature of the dashboard communicates these complex statistical concepts, allowing users to intuitively grasp the mechanics of missingness bias. However, the framework is not without limitations. To ensure the interactive dashboard runs efficiently in a browser, the simulation loops utilize a simplified Random Intercept model assuming homoscedasticity, rather than the Unstructured Covariance matrix applied to the empirical TLC data. Furthermore, as noted previously, the simulation does not currently model MNAR mechanisms.

Future iterations of this software could seek to implement complex covariance structures within the simulation loops, and introduce MNAR parameters to test the limits of Maximum Likelihood estimation. Ultimately, this framework underscores a vital biostatistical principle: robust trial design and rigorous covariate adjustment are strictly necessary to preserve the mathematical integrity of clinical research.

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Primary Literature & Clinical Context

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Data Source

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Generative AI Tool

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Appendix

A. R Code for Analysis and Simulation

Below is the full R code used for data cleaning, modeling, and simulation:

```r
# GLOBAL SETTING: This hides all code in the main report by default
knitr::opts_chunk$set(
  echo = FALSE,
  warning = FALSE,
  message = FALSE,
  out.width = "70%",
  fig.align = "center"
)
library(tidyverse)
library(gtsummary)

lead_mock <- read_csv("tlc_mock.csv") %>%
  mutate(
    # Convert Group to factor with specific levels/labels
    Group = factor(Group, levels = c("P", "A"), labels = c("Placebo", "Succimer")),
    # Ensure ID is a factor
    ID = as.factor(ID)
  )

# Create the summary table
summary_table <- lead_mock %>%
  select(Group, sex, age, Week0, Week1, Week4, Week6) %>%
  tbl_summary(
    by = Group,
    label = list(
      sex = "Sex",
      age = "Age (months)",
      Week0 = "Week 0 (Baseline)",
      Week1 = "Week 1",
      Week4 = "Week 4",
      Week6 = "Week 6"
    ),
    statistic = list(all_continuous() ~ "{median} ({min} - {max})"),
    missing = "no"
  ) %>%
  add_overall() %>%
  add_p() %>%
  modify_header(
    label = "**Variable**",
    stat_1 = "**Active**\n(N={n})",
    stat_2 = "**Placebo**\n(N={n})",
    stat_0 = "**Overall**\n(N={N})",
    p.value = "**p-value**"
  ) %>%
  bold_labels() %>%
  modify_caption("**Blood Lead Level Summary by Treatment Group**")

# Print the final table
summary_table
```

```r
# Create a tibble for the dictionary
dictionary <- tibble::tribble(
  ~Variable, ~Type, ~Description,
  "ID", "Integer", "A unique identifier for each subject.",
  "Group", "Character", "The treatment group: 'A' (Active) or 'P' (Placebo).",
  "Week0", "Numeric", "Blood lead level at baseline (Week 0).",
  "Week1", "Numeric", "Blood lead level after one week.",
  "Week4", "Numeric", "Blood lead level after four weeks.",
  "Week6", "Numeric", "Blood lead level after six weeks.",
  "sex", "Character", "The biological sex of the subject.",
  "age", "Numeric", "The age of the subject in months."
)

# Print out table
knitr::kable(dictionary, caption = "**Data Dictionary for the Simulated TLC Dataset**")
# 1. Reshape Data (Wide to Long) using lead_mock
lead_long <- lead_mock %>%
  pivot_longer(
    cols = c(Week0, Week1, Week4, Week6),
    names_to = "Week",
    values_to = "LeadLevel"
  ) %>%
  mutate(
    # Create a nice label for plotting
    Week_Label = factor(Week, levels = c("Week0", "Week1", "Week4", "Week6"),
                        labels = c("Week 0", "Week 1", "Week 4", "Week 6"))
  )

# Figure 1: Trends Over Time
p1 <- ggplot(lead_long, aes(x = Week_Label, y = LeadLevel, group = ID, color = Group)) +
  geom_line(alpha = 0.6) +
  labs(title = "Individual Blood Lead Levels Over Time",
       subtitle = "Active treatment group shows a sharp initial drop",
       x = "Time Point",
       y = "Blood Lead Level",
       color = "Treatment Group") +
  theme_minimal() +
  theme(legend.position = 'bottom')

p1
# Figure 2: Group Distributions
p2 <- ggplot(lead_long, aes(x = Week_Label, y = LeadLevel, fill = Group)) +
  geom_boxplot() +
  labs(title = "Distribution of Blood Lead Levels by Group",
       subtitle = "Active treatment group's median drops and stays low",
       x = "Time Point",
       y = "Blood Lead Level",
       fill = "Treatment Group") +
  theme_minimal() +
  theme(legend.position = 'bottom')

p2
# 2. Calculate Change Score
```

```r
lead_change <- lead_mock %>%
  mutate(LeadChange = Week6 - Week0)

# Figure 3: Change Over Time Distributions
p3 <- ggplot(lead_change, aes(x = LeadChange, fill = Group)) +
  geom_density(alpha = 0.7) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(title = "Distribution of Total Lead Change (Week 6 - Week 0)",
       subtitle = "Active group shows sharp decrease; Placebo shows slight decrease",
       x = "Change in Blood Lead Level",
       y = "Density",
       fill = "Treatment Group") +
  theme_minimal() +
  theme(legend.position = 'bottom')

p3
library(tidyverse)
library(nlme)
library(emmeans)

# --- 1. Prepare Data for Plot 1 (Observed) ---
lead_summary_plot <- lead_mock %>%
  pivot_longer(
    cols = c(Week0, Week1, Week4, Week6),
    names_to = "Week",
    values_to = "LeadLevel"
  ) %>%
  mutate(
    Time_Week = as.numeric(str_extract(Week, "[0-9]+")),
    Group = factor(Group, levels = c("Placebo", "Succimer"))
  ) %>%
  group_by(Time_Week, Group) %>%
  summarise(
    Mean_Lead = mean(LeadLevel),
    SEM_Lead = sd(LeadLevel) / sqrt(n()),
    .groups = 'drop'
  )

# --- 2. Prepare Data for Plot 2 (Predicted) ---
lead_long_nlme <- lead_mock %>%
  pivot_longer(
    cols = c(Week0, Week1, Week4, Week6),
    names_to = "Week",
    values_to = "LeadLevel"
  ) %>%
  mutate(
    Week = factor(Week, levels = c("Week0", "Week1", "Week4", "Week6"))
  )

# FIT THE ADJUSTED MODEL (Matching Dashboard)
model_nlme <- lme(
  LeadLevel ~ Group * Week + age + sex,
  random = ~ 1 | ID,
```

```r
  weights = varIdent(form = ~ 1 | Week),
  correlation = corSymm(form = ~ 1 | ID),
  data = lead_long_nlme,
  control = lmeControl(maxIter = 10000, msMaxIter = 10000)
)

# Marginal Means (Averaged over Age and Sex)
model_predictions_emmeans <- emmeans(model_nlme, ~ Group | Week) %>%
  as.data.frame()

model_predictions_scaled <- model_predictions_emmeans %>%
  mutate(
    Time_Week_Numeric = as.numeric(str_extract(Week, "[0-9]+"))
  )

# --- 3. Plotting Logic ---
y_min_obs <- min(lead_summary_plot$Mean_Lead - lead_summary_plot$SEM_Lead)
y_max_obs <- max(lead_summary_plot$Mean_Lead + lead_summary_plot$SEM_Lead)
y_min_pred <- min(model_predictions_scaled$lower.CL)
y_max_pred <- max(model_predictions_scaled$upper.CL)
global_y_min <- floor(min(y_min_obs, y_min_pred))
global_y_max <- ceiling(max(y_max_obs, y_max_pred))

# Plot 1: Observed
p_observed <- ggplot(lead_summary_plot, aes(x = Time_Week, y = Mean_Lead, group = Group, color = Group)
  geom_line(linewidth = 0.8) +
  geom_point(aes(shape = Group), size = 3) +
  geom_errorbar(aes(ymin = Mean_Lead - SEM_Lead, ymax = Mean_Lead + SEM_Lead), width = 0.2, alpha = 0.7
  labs(title = "Observed Mean Blood Lead Levels", subtitle = "Raw averages (Unadjusted)",
       x = "Time (weeks)", y = "Mean blood lead level (mcg/dL)", color = "Treatment", shape = "Treatmen
  scale_x_continuous(breaks = c(0, 1, 4, 6)) +
  ylim(global_y_min, global_y_max) + theme_minimal() + theme(legend.position = "bottom")

p_observed
# Plot 2: Predicted
p_predicted <- ggplot(model_predictions_scaled, aes(x = Time_Week_Numeric, y = emmean, group = Group, c
  geom_line(linewidth = 0.8) +
  geom_point(aes(shape = Group), size = 3) +
  geom_ribbon(aes(ymin = lower.CL, ymax = upper.CL, fill = Group), alpha = 0.2, color = NA) +
  labs(title = "Predicted Mean Blood Lead Levels", subtitle = "Adjusted for Age and Sex",
       x = "Time (weeks)", y = "Predicted mean blood lead level (mcg/dL)", color = "Treatment", shape =
  scale_x_continuous(breaks = c(0, 1, 4, 6)) +
  ylim(global_y_min, global_y_max) + theme_minimal() + theme(legend.position = "bottom")

p_predicted
library(tidyverse)
library(nlme)

# 1. Prepare data for the nlme model
lead_mixed_nlme <- lead_mock %>%
  pivot_longer(
    cols = c(Week0, Week1, Week4, Week6),
    names_to = "Week",
```

```r
    values_to = "LeadLevel"
  ) %>%
  mutate(
    Week = factor(Week, levels = c("Week0", "Week1", "Week4", "Week6")),
    # Group is already a factor from the Data Summary chunk
    ID = as.factor(ID)
  )

# 2. Fit the ADJUSTED mixed-effects model
model_nlme <- lme(
  LeadLevel ~ Group * Week + age + sex,
  random = ~ 1 | ID,
  weights = varIdent(form = ~ 1 | Week),
  correlation = corSymm(form = ~ 1 | ID),
  data = lead_mixed_nlme,
  control = lmeControl(maxIter = 10000, msMaxIter = 10000)
)

# 3. Format Summary as a clean table
fixed_effects <- as.data.frame(summary(model_nlme)$tTable)
# Rename the rows so they are highly readable for a layman/reviewer
rownames(fixed_effects) <- c("Intercept (Placebo, Wk 0)", "Treatment: Succimer",
                             "Time: Week 1", "Time: Week 4", "Time: Week 6",
                             "Age (Months)", "Sex: Male",
                             "Succimer x Week 1", "Succimer x Week 4", "Succimer x Week 6")

knitr::kable(fixed_effects, digits = 3, caption = "**Fixed Effects from the Adjusted Mixed-Effects Model

# --- EXTRACT VALUES FOR DYNAMIC TEXT (PART 1) ---
base_diff_p <- summary(model_nlme)$tTable["GroupSuccimer", "p-value"]
wk1_int_p   <- summary(model_nlme)$tTable["GroupSuccimer:WeekWeek1", "p-value"]
wk4_int_p   <- summary(model_nlme)$tTable["GroupSuccimer:WeekWeek4", "p-value"]
wk6_int_p   <- summary(model_nlme)$tTable["GroupSuccimer:WeekWeek6", "p-value"]

# Helper function to prevent scientific notation and format small p-values
format_p <- function(p) {
  if (p < 0.001) return("< 0.001")
  else return(paste0("= ", round(p, 3)))
}
# 4. Perform the post-hoc contrast using emmeans
library(emmeans)

# Calculate estimated marginal means
emm_week <- emmeans(model_nlme, ~ Group | Week)

# Calculate pairwise comparisons (Succimer - Placebo)
contrasts_all <- pairs(emm_week, reverse = TRUE)

# Calculate Confidence Intervals (95%)
ci_all <- confint(contrasts_all)

# Convert to data frames for easy handling
contrasts_df <- as.data.frame(contrasts_all)
```

```r
ci_df        <- as.data.frame(ci_all)

# Filter for Week 6
contrast_wk6_row <- contrasts_df[contrasts_df$Week == "Week6", ]
ci_wk6_row       <- ci_df[ci_df$Week == "Week6", ]

# --- CREATE PRESENTATION TABLE ---
# Stitching the estimate, CI, and p-value into a clean data frame
wk6_table_display <- data.frame(
  Contrast = "Succimer vs. Placebo",
  Estimate = contrast_wk6_row$estimate,
  SE = contrast_wk6_row$SE,
  `Lower 95% CI` = ci_wk6_row$lower.CL,
  `Upper 95% CI` = ci_wk6_row$upper.CL,
  `p-value` = contrast_wk6_row$p.value,
  check.names = FALSE # Keeps the spaces in column names
)

knitr::kable(wk6_table_display, digits = 3, caption = "**Post-Hoc Treatment Contrast at Week 6**")

# --- EXTRACT VALUES FOR DYNAMIC TEXT (PART 2) ---
wk6_diff_est <- abs(contrast_wk6_row$estimate) # Magnitude (positive number)
wk6_diff_p   <- contrast_wk6_row$p.value
wk6_lower    <- ci_wk6_row$lower.CL
wk6_upper    <- ci_wk6_row$upper.CL
library(tidyverse)
library(nlme)
library(foreach)
library(doParallel)

# --- 1. DEFINE SIMULATION FUNCTIONS ---

# MACHINE 1: Data Generator
generate_trial_data <- function(n_subjects = 100) {
  subject_id <- 1:n_subjects
  group <- rep(c("Placebo", "Succimer"), each = n_subjects / 2)
  age <- runif(n_subjects, min = 6.5, max = 38.6)
  age_c <- age - 22
  b_0i <- rnorm(n_subjects, mean = 0, sd = 4.42134)

  subjects <- data.frame(ID = subject_id, Group = group, age = age, age_c = age_c, b_0i = b_0i)
  long_data <- expand.grid(Week = c("Week0", "Week1", "Week4", "Week6"), ID = subject_id)
  long_data <- merge(long_data, subjects, by = "ID")

  long_data$LeadLevel <- NA
  for (i in 1:nrow(long_data)) {
    grp <- long_data$Group[i]
    wk <- as.character(long_data$Week[i])
    ac <- long_data$age_c[i]
    b_0 <- long_data$b_0i[i]

    expected_val <- 26.4
    if (grp == "Placebo") {
```

```r
      if (wk == "Week1") expected_val <- 24.7
      if (wk == "Week4") expected_val <- 24.1
      if (wk == "Week6") expected_val <- 23.6
    } else if (grp == "Succimer") {
      if (wk == "Week1") expected_val <- 13.5
      if (wk == "Week4") expected_val <- 15.5
      if (wk == "Week6") expected_val <- 20.8
    }

    expected_val <- expected_val - (0.5 * ac)

    epsilon <- rnorm(1, mean = 0, sd = 2.460057)
    long_data$LeadLevel[i] <- expected_val + b_0 + epsilon
  }

  long_data$Week <- factor(long_data$Week, levels = c("Week0", "Week1", "Week4", "Week6"))
  long_data$Group <- factor(long_data$Group, levels = c("Placebo", "Succimer"))
  long_data$ID <- as.factor(long_data$ID)
  return(long_data)
}

# MACHINE 2A: MCAR Mechanism
apply_mcar <- function(data, missing_prop = 0.20) {
  n_delete <- floor(nrow(data) * missing_prop)
  eligible_indices <- which(data$Week != "Week0")
  delete_indices <- sample(eligible_indices, n_delete)
  data$LeadLevel[delete_indices] <- NA
  return(data)
}

# --- 2. RUN MCAR SIMULATION ---
n_sims <- 1000
true_est_sim <- -2.8

num_cores <- parallel::detectCores() - 1
my_cluster <- makeCluster(num_cores)
registerDoParallel(my_cluster)
clusterSetRNGStream(my_cluster, 123)

sim_results_mcar <- foreach(i = 1:n_sims, .combine = rbind, .packages = c("nlme")) %dopar% {
  clean_data <- generate_trial_data(n_subjects = 100)
  test_data <- apply_mcar(clean_data, missing_prop = 0.20)

  tryCatch({
    sim_model <- lme(LeadLevel ~ Group * Week, random = ~ 1 | ID, data = test_data, na.action = na.exclu
    est <- summary(sim_model)$tTable["GroupSuccimer:WeekWeek6", "Value"]
    se <- summary(sim_model)$tTable["GroupSuccimer:WeekWeek6", "Std.Error"]
    c(Estimate = est, SE = se)
  }, error = function(e) c(Estimate = NA, SE = NA))
}
stopCluster(my_cluster)

results_df_mcar <- na.omit(as.data.frame(sim_results_mcar))
```

```r
# Calculate Coverage and Table Metrics
results_df_mcar <- results_df_mcar %>%
  mutate(
    Lower_CI = Estimate - 1.96 * SE,
    Upper_CI = Estimate + 1.96 * SE,
    Covers = ifelse(true_est_sim >= Lower_CI & true_est_sim <= Upper_CI, 1, 0)
  )

mean_sim_mcar <- mean(results_df_mcar$Estimate)
bias_mcar <- mean_sim_mcar - true_est_sim
sd_mcar <- sd(results_df_mcar$Estimate)
coverage_mcar <- mean(results_df_mcar$Covers) * 100
ggplot(results_df_mcar, aes(x = Estimate)) +
  geom_histogram(bins = 30, fill = "steelblue", alpha = 0.7, color = "white") +
  geom_vline(aes(xintercept = true_est_sim, color = "True Estimate"), linewidth = 1.2, linetype = "dash
  geom_vline(aes(xintercept = mean_sim_mcar, color = "Mean Simulated"), linewidth = 1.2) +
  labs(title = "Study 1: MCAR Results (1,000 Trials)", subtitle = "Estimates remain unbiased despite 20
  scale_color_manual(values = c("True Estimate" = "black", "Mean Simulated" = "red")) +
  theme_minimal() + theme(legend.position = "bottom")
table_mcar <- data.frame(
  Metric = c("Mean Treatment Effect", "Bias", "Empirical Standard Deviation", "CI Coverage (%)"),
  Value = c(mean_sim_mcar, bias_mcar, sd_mcar, coverage_mcar)
)
knitr::kable(table_mcar, digits = 3, caption = "**Simulation Statistics: MCAR**")
# MACHINE 2B: MAR Mechanism
apply_mar <- function(data, missing_prop = 0.20) {
  n_delete <- floor(nrow(data) * missing_prop)
  eligible_indices <- which(data$Week != "Week0")

  weights <- ifelse(data$Group[eligible_indices] == "Placebo",
                    (40 - data$age[eligible_indices])^3,
                    1)

  delete_indices <- sample(eligible_indices, n_delete, prob = weights)
  data$LeadLevel[delete_indices] <- NA
  return(data)
}

# --- RUN MAR SIMULATION ---
my_cluster <- makeCluster(num_cores)
registerDoParallel(my_cluster)
clusterSetRNGStream(my_cluster, 456)

sim_results_mar <- foreach(i = 1:n_sims, .combine = rbind, .packages = c("nlme")) %dopar% {
  clean_data <- generate_trial_data(n_subjects = 100)
  test_data <- apply_mar(clean_data, missing_prop = 0.20)

  # MODEL A: UNADJUSTED
  est_A <- NA; se_A <- NA
  try({
    m1 <- lme(LeadLevel ~ Group * Week, random = ~1|ID, data=test_data, na.action=na.exclude, control=l
    est_A <- summary(m1)$tTable["GroupSuccimer:WeekWeek6", "Value"]
    se_A <- summary(m1)$tTable["GroupSuccimer:WeekWeek6", "Std.Error"]
```

```
  }, silent=TRUE)

  # MODEL B: ADJUSTED
  est_B <- NA; se_B <- NA
  try({
    m2 <- lme(LeadLevel ~ Group * Week + age_c, random = ~1|ID, data=test_data, na.action=na.exclude, co
    est_B <- summary(m2)$tTable["GroupSuccimer:WeekWeek6", "Value"]
    se_B <- summary(m2)$tTable["GroupSuccimer:WeekWeek6", "Std.Error"]
  }, silent=TRUE)

  c(est_A, se_A, est_B, se_B)
}
stopCluster(my_cluster)

results_df_mar <- na.omit(as.data.frame(sim_results_mar))
colnames(results_df_mar) <- c("Unadjusted", "Unadj_SE", "Adjusted", "Adj_SE")

# Calculate Coverage and Table Metrics
results_df_mar <- results_df_mar %>%
  mutate(
    Unadj_Lower = Unadjusted - 1.96 * Unadj_SE,
    Unadj_Upper = Unadjusted + 1.96 * Unadj_SE,
    Unadj_Covers = ifelse(true_est_sim >= Unadj_Lower & true_est_sim <= Unadj_Upper, 1, 0),

    Adj_Lower = Adjusted - 1.96 * Adj_SE,
    Adj_Upper = Adjusted + 1.96 * Adj_SE,
    Adj_Covers = ifelse(true_est_sim >= Adj_Lower & true_est_sim <= Adj_Upper, 1, 0)
  )

mean_unadj <- mean(results_df_mar$Unadjusted)
mean_adj   <- mean(results_df_mar$Adjusted)
bias_unadj <- mean_unadj - true_est_sim
bias_adj   <- mean_adj - true_est_sim
ggplot(results_df_mar, aes(x = Unadjusted)) +
  geom_histogram(bins = 30, fill = "#D9534F", alpha = 0.8, color = "white") +
  geom_vline(aes(xintercept = true_est_sim, color = "True Estimate"), linewidth = 1.2, linetype = "dashe
  geom_vline(aes(xintercept = mean_unadj, color = "Mean Simulated"), linewidth = 1.2) +
  labs(title = "Study 2A: Unadjusted Model Results (High Bias)", subtitle = "Ignoring Age causes severe
  scale_color_manual(values = c("True Estimate" = "black", "Mean Simulated" = "darkred")) +
  theme_minimal() + theme(legend.position = "bottom")
ggplot(results_df_mar, aes(x = Adjusted)) +
  geom_histogram(bins = 30, fill = "steelblue", alpha = 0.8, color = "white") +
  geom_vline(aes(xintercept = true_est_sim, color = "True Estimate"), linewidth = 1.2, linetype = "dashe
  geom_vline(aes(xintercept = mean_adj, color = "Mean Simulated"), linewidth = 1.2) +
  labs(title = "Study 2B: Adjusted Model Results (Low Bias)", subtitle = "Correcting for Age restores th
  scale_color_manual(values = c("True Estimate" = "black", "Mean Simulated" = "red")) +
  theme_minimal() + theme(legend.position = "bottom")
table_mar <- data.frame(
  Model = c("Unadjusted (Naive)", "Adjusted (Corrected)"),
  `Mean Estimate` = c(mean_unadj, mean_adj),
  Bias = c(bias_unadj, bias_adj),
  `Empirical SD` = c(sd(results_df_mar$Unadjusted), sd(results_df_mar$Adjusted)),
  `Coverage (%)` = c(mean(results_df_mar$Unadj_Covers)*100, mean(results_df_mar$Adj_Covers)*100),
```

```
  check.names = FALSE
)
knitr::kable(table_mar, digits = 3, caption = "**Simulation Statistics: MAR (Differential Dropout)**")
# Q-Q plot for lme objects
qqnorm(model_nlme, abline = c(0, 1), main = "Q-Q Plot of Standardized Residuals")
# Standard residual plot for lme objects
plot(model_nlme, main = "Standardized Residuals vs. Fitted Values")
```

B. Model Diagnostics

The following plots were used to verify the assumptions of the linear mixed-effects model.

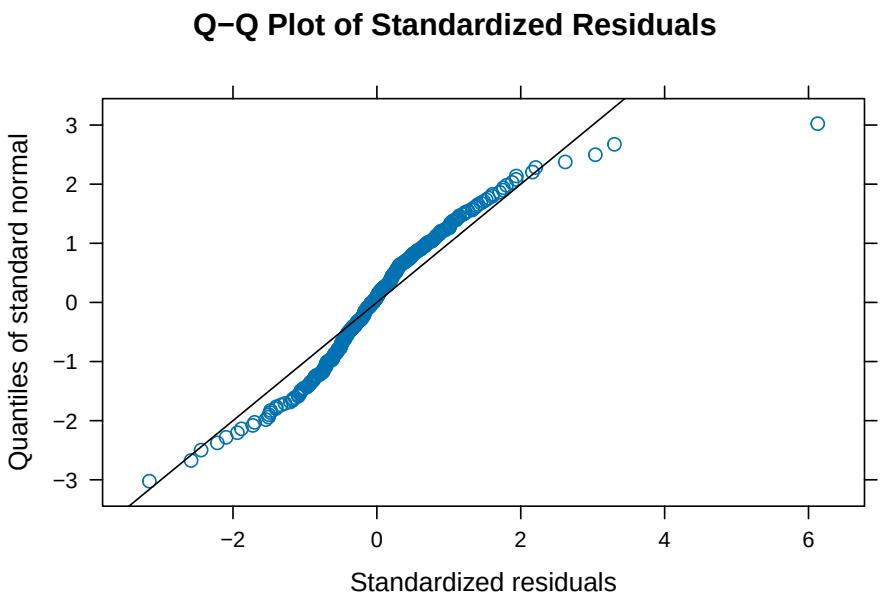

Figure 9. Q-Q Plot of Standardized Residuals. Diagnostic plot verifying the normality assumption of the residual error.

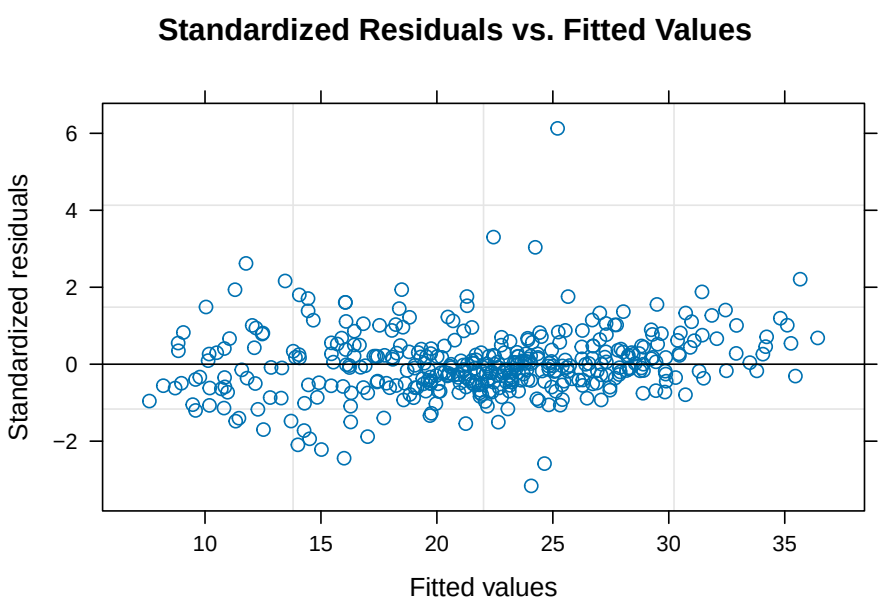

Figure 10. Standardized Residuals vs. Fitted Values. Diagnostic plot verifying the assumption of homoscedasticity and the adequacy of the variance structure.